

Lab-7 (Pointer)

1. Write a c-program to show basic declaration of pointer.

Ans:

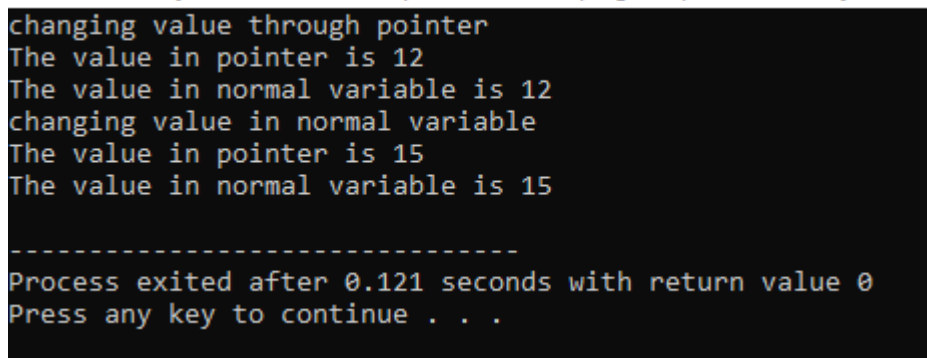
```
#include<stdio.h>
int main()
{
    int a;
    int *p;
    p=&a;
    return 0;
}
```

2. Write a c-program to demonstrate the use of (&) and (*) operator.

Ans:

```
#include<stdio.h>
int main()
{
    int a;
    int *p;
    p=&a; //linking pointer to normal variable
    *p=12; //changing value through pointer
    printf("changing value through pointer\n");
    printf("The value in pointer is %d\n",*p);
    printf("The value in normal variable is %d\n",a);
    a=15; //changing value in normal variable
    printf("changing value in normal variable\n");
    printf("The value in pointer is %d\n",*p);
    printf("The value in normal variable is %d\n",a);
    return 0;
}
```

Output:  C:\Users\ayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe



```
changing value through pointer
The value in pointer is 12
The value in normal variable is 12
changing value in normal variable
The value in pointer is 15
The value in normal variable is 15

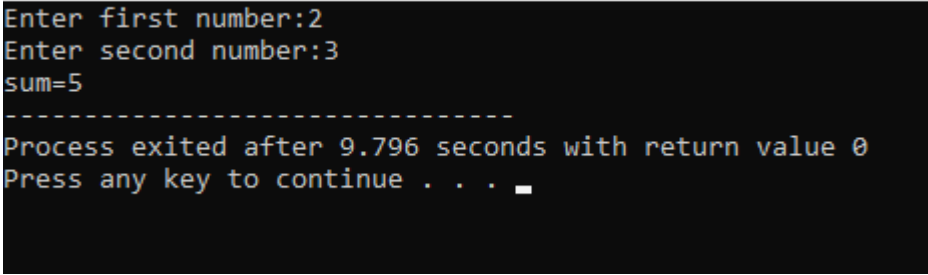
-----
Process exited after 0.121 seconds with return value 0
Press any key to continue . . .
```

3. Write a c-program to add two number using pointer.

Ans:

```
#include<stdio.h>
int main()
{
    int a,b,sum;
    int *x=&a,*y=&b;
    printf("Enter first number:");
    scanf("%d",x);
    printf("Enter second number:");
    scanf("%d",y);
    sum=*x+*y;
    printf("sum=%d",sum);
    return 0;
}
```

Output:  C:\Users\ayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe



```
Enter first number:2
Enter second number:3
sum=5
-----
Process exited after 9.796 seconds with return value 0
Press any key to continue . . .
```

4. Write a c-program to swap two number using call by reference.

Ans:

```
#include<stdio.h>
int main()
{
    int a,b,*x=&a,*y=&b;
    printf("Enter first number:");
    scanf("%d",x);
    printf("Enter second number:");
    scanf("%d",y);
    *x=*x+*y;
    *y=*x-*y;
    *x=*x-*y;
    printf("The value after swaping are: %d, %d",*x,*y);
    return 0;
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```
Enter first number:2
Enter second number:3
The value after swaping are: 3, 2
-----
Process exited after 2.811 seconds with return value 0
Press any key to continue . . .
```

5. Write a c-program to find maximum among two number using pointer.

Ans:

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int a,b,*x=&a,*y=&b;
```

```
    printf("Enter first number:");
```

```
    scanf("%d",x);
```

```
    printf("Enter second number:");
```

```
    scanf("%d",y);
```

```
    if(*x>*y)
```

```
        printf("%d is greatest number",*x);
```

```
    else
```

```
        printf("%d is greatest number",*y);
```

```
    return 0;
```

```
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```
Enter first number:3
Enter second number:9
9 is greatest number
-----
Process exited after 11.99 seconds with return value 0
Press any key to continue . . .
```

6. Write a c-program to find factorial of number using recursion function and pointer.

Ans:

```
#include<stdio.h>
```

```
int fact(int *x);
```

```
int main()
```

```
{
```

```
    int n,*x;
```

```
    x=&n;
```

```
    printf("Enter any number:");
```

```
    scanf("%d",x);
```

```
    if(*x<0)
```

```
        printf("There is no factorial for negative number");
```

```

        else
        {
            printf("The factorial of %d is:%d",*x,fact(x));
        }
        return 0;

    }
    int fact(int *x)
    {
        int a=*x-1;
        if(*x>1)
        {
            return *x * fact(&a);
        }
        else
        {
            return 1;
        }
    }
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```

Enter any number:6
The factorial of 6 is:720
-----
Process exited after 9.359 seconds with return value 0
Press any key to continue . . .

```

7. Write a c-program to input array element and print them using pointer.

Ans:

```

#include<stdio.h>
int main()
{
    int size;
    printf("Enter size of array:");
    scanf("%d",&size);
    int i,a[size];
    printf("Enter element for an array:");
    for(i=0; i<size; i++)
    {
        scanf("%d",(a+i));
    }

    printf("The inputed element are:");
    for(i=0; i<size; i++)

```

```

    {
        printf("%d,",*(a+i));
    }
    return 0;
}

```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```

Enter size of array:5
Enter element for an array:1
2
3
4
5
The inputted element are:1,2,3,4,5,
-----
Process exited after 8.256 seconds with return value 0
Press any key to continue . . .

```

8. Write a c-program to find minimum element of array using pointer.

Ans:

```

#include<stdio.h>
int main()
{
    int size;
    printf("Enter size of array:");
    scanf("%d",&size);
    int a[size];
    int i,min;

    printf("Enter an element:");
    for(i=0; i<size; i++)
    {
        scanf("%d",&a[i]);
    }

    min=a[0];
    for(i=0; i<size; i++)
    {
        if(a[i]<min)
        {
            min=a[i];
        }
    }
    printf("The minimum element of array is %d",min);
}

```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```
Enter size of array:5
Enter an element:2
9
3
7
1
The minimum element of array is 1
-----
Process exited after 13.7 seconds with return value 33
Press any key to continue . . .
```

9. Write a c-program to demonstrate pointer arithmetic operation.

Ans:

```
#include<stdio.h>
int main()
{
    int a=2,c=4;
    double b=3;
    double *y;
    int *x,*z;
    x=&a;
    y=&b;
    z=&c;

    printf("The value of a and x pointer=%d,%d\n",a,*x);
    printf("The address of normal variable a=%d\n",&a);
    printf("The address of pointer x=%d\n",x);
    x=x+1;
    printf("The address of pointer X after operatio=%d\n",x);
    printf("The value of a and x pointer after opreation=%d,%d\n\n",a,*x);

    printf("The value of b and y pointer=%d,%f\n",b,*y);
    printf("The address of normal variable b=%f\n",&b);
    printf("The address of pointer y=%f\n",y);
    y=y-1;
    printf("The address of pointer y after operatio=%f\n",y);
    printf("The value of b and y pointer after opreation=%f,%f\n",b,*y);

}
```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```
The value of a and x pointer=2,2
The address of normal variable a=6487556
The address of pointer x=6487556
The address of pointer X after operatio=6487560
The value of a and x pointer after opreation=2,6487552

The value of b and y pointer=0,3.000000
The address of normal variable b=0.000000
The address of pointer y=0.000000
The address of pointer y after operatio=0.000000
The value of b and y pointer after opreation=3.000000,0.000000

-----
Process exited after 0.09119 seconds with return value 63
Press any key to continue . . .
```

10. Write a c-program to do operation using pointer chain.

Ans:

```
#include<stdio.h>
int main()
{
    int a=5;
    int *x;
    int **y;
    int ***z;
    x=&a;
    y=&x;
    z=&y;

    printf("The value of a=%d\n",a);
    *x=*x+1;
    printf("The value of pointer x=%d\n",*x);
    **y=*x+2;
    printf("The value of pointer y=%d\n",**y);
    ***z=**y+3;
    printf("The value of pointer z=%d\n",***z);
    return 0;
}
```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```
The value of a=5
The value of pointer x=6
The value of pointer y=8
The value of pointer z=11

-----
Process exited after 0.102 seconds with return value 0
Press any key to continue . . .
```

11. write a c-program to input 2d array element and print them using pointer.

Ans:

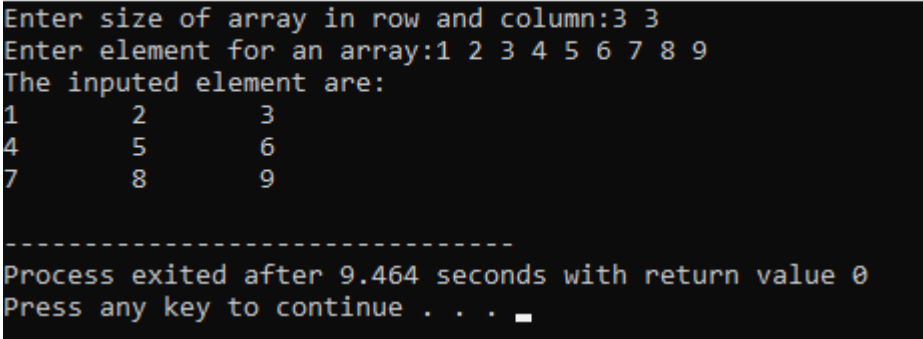
```
#include<stdio.h>
int main()
{
    int row,col;
    int i,j;
    printf("Enter size of array in row and column:");
    scanf("%d%d",&row,&col);
    int a[row][col];
    printf("Enter element for an array:");
    for(i=0; i<row; i++)
    {
        for(j=0; j<col; j++)
        {

            scanf("%d",&a[i][j]);
        }
    }

    printf("The inputted element are:\n");
    for(i=0; i<row; i++)
    {
        for(j=0; j<col; j++)
        {

            printf("%d\t",&a[i][j]);
        }
    }
    printf("\n");
    return 0;
}
```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe



```
Enter size of array in row and column:3 3
Enter element for an array:1 2 3 4 5 6 7 8 9
The inputted element are:
1      2      3
4      5      6
7      8      9

-----
Process exited after 9.464 seconds with return value 0
Press any key to continue . . .
```


12. Write a c-program to find maximum element of each row and column of 2d array with use of pointer.

Ans:

```
#include<stdio.h>
int main()
{
    int row,col;
    int i,j;
    printf("Enter size of array in row and column:");
    scanf("%d%d",&row,&col);
    int a[row][col];
    printf("Enter element for an array:");

    for(i=0; i<row; i++)
    {
        for(j=0; j<col; j++)
        {
            scanf("%d",&a[i][j]);
        }
    }

    int max,k=0;
    for(i=0; i<row; i++)
    {
        max=a[i][0];
        for(j=0; j<col; j++)
        {
            if(a[i][j]>max)
            {
                max=a[i][j];
            }
        }
        printf("The maximum element of %d row is:%d\n",i,max);
    }

    for(j=0; j<col; j++)
    {
        max=a[0][j];
        for(i=0; i<row; i++)
        {
            if(a[i][j]>max)
            {
```

```

        max=a[i][j];
    }

    }

    printf("The maximum element of %d column is:%d\n",j,max);
}
return 0;
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```

Enter size of array in row and column:3 3
Enter element for an array:1 2 3 4 5 6 7 8 9
The maximum element of 0 row is:3
The maximum element of 1 row is:6
The maximum element of 2 row is:9
The maximum element of 0 column is:7
The maximum element of 1 column is:8
The maximum element of 2 column is:9

-----
Process exited after 18.13 seconds with return value 0
Press any key to continue . . .

```

13. Write a c-program to demonstrate use of array of pointer.

Ans:

```

#include<stdio.h>
int main()
{
    int a=1,b=2,c=3,d=4;
    int *p[4];
    p[0]=&a;
    p[1]=&b;
    p[2]=&c;
    p[3]=&d;
    int i;
    printf("The element of an array are:\n");
    for(i=0; i<4; i++)
    {

        printf("%d,",*p[i]);

    }
    return 0;
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe

```

The element of an array are:
1,2,3,4,
-----
Process exited after 0.09885 seconds with return value 0
Press any key to continue . . .

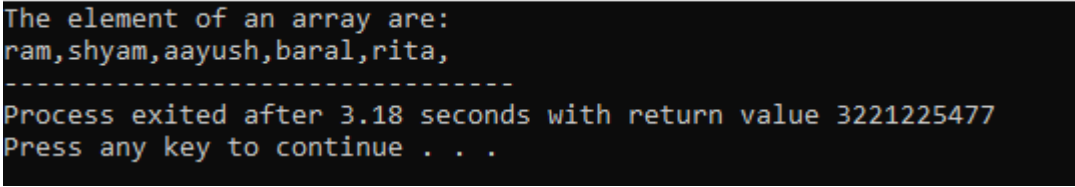
```

14. Write a c-program to demonstrate use of array of pointer.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
    char *p[]={"ram","shyam","aayush","baral","rita"};
    int n=sizeof(p);
    int i;
    printf("The element of an array are:\n");
    for(i=0; i<n; i++)
    {
        printf("%s,",p[i]);
    }
    return 0;
}
```

Output:  C:\Users\ayus\OneDrive\Desktop\New folder\C-program\pointer next day.exe



```
The element of an array are:
ram,shyam,aayush,baral,rita,
-----
Process exited after 3.18 seconds with return value 3221225477
Press any key to continue . . .
```